

Nikhil R. Agrawal - Curriculum Vitae

Process Engineer (H1B Visa holder)

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Deposition Product Group, Lam Research

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EDUCATION

Ph.D. in Chemical Engineering

August 2024

University of California, Berkeley, USA

Specialization area: Statistical physics, Electrical Double Layers, Electrolytes, Chemical Physics

Thesis title: *Modified Gaussian Renormalized Fluctuation Theory for Electrolytes at Interfaces*

Advisor: Prof. Rui Wang

B. Tech. and M. Tech (Dual Degree) in Chemical Engineering

July 2018

Indian Institute of Technology (IIT), Delhi, India

Institute Silver Medal for highest GPA in the program

Specialization area: Multiphase Reactors, Techno-economic analysis of processes, Mathematical Modeling

Master's thesis title: *Microscopic characterization of particle packings in packed bed reactors*

Master's thesis advisor: Prof. Shantanu Roy

WORK EXPERIENCE

Designing PEALD and PECVD processes for dielectric film materials

Oct '24 - Present

Process Engineer | PEALD Deposition Product Group, Lam Research, Oregon, USA

Developed next generation low k (<3.5) ALD Liner for Through Silicon Vias (*patent pending*) and solved the defects problem for oxide-nitride stack in Lam's Striker chambers. Other contributions include:

- Hardware component evaluation to improve throughput for Lam's Striker Chambers
- Evaluated multiple ALD precursors to developed carbon and fluorine doped ALD Oxide films, gaining experience in crucial organic chemical reactions in semiconductor processing
- Coordinating and driving a multinational team of software, mechanical, electrical, and RF plasma engineers, together with Lam's Field Service Engineers in Taiwan, to ensure smooth project execution
- Developed a PECVD SIN diffusion barrier film, also worked on ALD Nitride films for same application

Statistical physics of electrical double layers in concentrated electrolytes

Aug '18 - Aug '24

Ph.D. Candidate | Chemical Engineering, University of California, Berkeley, USA

Original research work to explain how highly concentrated salt solutions behave near charged surfaces, leading to 7 first author research publications and multiple conference presentations. Key contributions include:

- Developed a new theoretical model to accurately model short and long range electrostatic correlation in electrolytes
- Explained counterintuitive phenomena in systems with multivalent salts, including charge inversion, like-charge attraction, and opposite-charge repulsion
- Developed a new spectral method based algorithm to efficiently calculate two point electrostatic correlation operator

Advanced ML and Bayesian Modeling for quantifying uncertainty in material models Mar '24 - June '24
Semester Intern | Physics Division, Lawrence Livermore National Laboratory, USA

Collaborated with a 10+ member multidisciplinary team to deploy Bayesian and deep learning methods for quantifying uncertainty in materials simulations, developing practical experience in AI/ML methods applicable across domains.

Data Science, Cloud Computing, and Monte-Carlo simulations for complex systems June '23 - Aug '23
Summer Intern | Center of Computational Biology, Flatiron Institute, New York, USA

Conducted large-scale hybrid Monte-Carlo and Molecular dynamics simulations to model protein behavior during cell division, building capabilities in Monte-Carlo methods, cloud-based computation, and scientific data engineering.

Reactor Catalyst-Packing optimization for efficiency gains in refining operations Jan '17 - July '18
Master's Candidate | Department of Chemical Engineering, IIT Delhi, India

Designed and validated a mathematical model describing catalyst-bed microstructure, providing insights that could improve packing-methods and led to 2 publications.

Large scale computer simulations to optimize catalyst packing in packed bed reactors May '17 - July '17
Visiting Researcher | Mentor: Prof. Dr. Thorsten Pöschel, University of Erlangen-Nuremberg, Germany

Part of a multinational India-Germany research team to optimize catalyst-packing in packed-bed reactors for improved crude-oil refining efficiency, applied cloud computing and Monte Carlo simulations to evaluate and refine packing strategies.

Image processing and pore-network modeling of nuclear reactors May '16 - July '16
Summer Intern | Porous Media Group, Institut de Mecanique des Fluides de Toulouse (IMFT), France

Worked at one of the world's largest fluid mechanics institutes and developed a software tool to analyze experimental images, enabling faster optimization of reactor design.

Design of solar-powered photocatalysts for conversion of CO₂ into commodity chemicals May '15 - April '16
Undergraduate Researcher | Mentor: Prof. Suddhasatwa Basu, Chemical Engineering, IIT Delhi, India

Photo-catalysis is the use of semiconducting materials and solar energy-generated electrons to drive chemical reactions, a principle shared with photovoltaic devices. In this project, I

- synthesized and characterized structure of semiconductor-graphene/silica nanocomposite catalysts, and
- analyzed the photo-catalytic activity of these catalysts through dye degradation in a *self-designed reactor*.

JOURNAL PUBLICATIONS

On the topic of electrolysis, electrolytes, and polyelectrolytes:

1. Electrostatic Correlation Augmented Self-Consistent Field Theory and Its Application to Polyelectrolyte Brushes
Chao Duan, Nikhil R. Agrawal, and Rui Wang, *Physical Review Letters* (2025) 134, 048101
2. Understanding long-range opposite-charge repulsion in multivalent salt solutions
Nikhil R. Agrawal, Carlo Carraro and Rui Wang, *J. Chem. Phys.* 161, 204902 (2024)
3. Nature of overcharging and charge inversion in electrical double layers
Nikhil R. Agrawal, Chao Duan, and Rui Wang, *J. Phys. Chem. B* 2024, 128, 1, 303–311
4. Ion correlation-driven like-charge attraction in multivalent salt solutions
Nikhil R. Agrawal, Ravtej Kaur, Carlo Carraro and Rui Wang, *J. Chem. Phys.* 159, 244905 (2023)
5. Non-monotonic salt concentration dependence of inverted electrokinetic flow
Nikhil R. Agrawal and Rui Wang, *AIChE Journal*, e18269, 2023.

6. Self-consistent description of vapor-liquid interface in ionic fluids
Nikhil R. Agrawal and Rui Wang, *Physical Review Letters* (2022), 129, 228001.
7. Electrostatic correlation induced ion condensation and charge inversion in multivalent electrolytes
Nikhil R. Agrawal and Rui Wang, *Journal of Chemical Theory and Computation* (2022), 18, 6271-6280
8. Sturm–Liouville theory inspired method to solve the Modified Gaussian Renormalized Fluctuation theory for electrolytes
Nikhil R. Agrawal, Carlo Carraro and Rui Wang, *in preparation*.

On the topic of trickle bed reactors, multiphase reactors, and process intensification:

1. A first-order segregation phenomenon in fluid-immersed granular systems
 Prapanch Nair, LAT Cisneros, CRK Windows-Yule, Nikhil R. Agrawal, Shantanu Roy, and Thorsten Pöschel, *Powder Technology* 373 (2020): 357-361.
2. Isotropy of sphere packings in a cylindrical confinement
Nikhil R. Agrawal, Prapanch Nair, Thorsten Pöschel and Shantanu Roy, *Chemical Engineering Journal* 377 (2019): 119820.

SELECTED PRESENTATIONS

1. Modified Gaussian Renormalized Fluctuation Theory for Electrolytes at Interfaces
 2025 AIChE Annual Meeting, Oral Presentation
Nikhil R. Agrawal and Rui Wang
2. Thermodynamics and transport of concentrated electrolytes at interfaces
 2025 AIChE Annual Meeting, Poster Presentation
Nikhil R. Agrawal and Rui Wang
3. Ion correlation induced non-monotonic height change and microphase separation of polyelectrolyte brushes
 2024 American Physical Society March Meeting, Oral Presentation
 Rui Wang, Chao Duan, Takashi Yokokura, and Nikhil R. Agrawal
4. Beyond mean-field Poisson-Boltzmann: A self-consistent theory for electrical double layers
 Energy Conversion Group, Lawrence Berkeley National Laboratory, Invited Presentation, Feb 2023
Nikhil R. Agrawal and Rui Wang
5. Modified Gaussian Renormalized Fluctuation theory: A self-consistent electrolyte solution theory
 Pitzer Center for Theoretical Chemistry, UC Berkeley, Invited Presentation, Jan 2023
Nikhil R. Agrawal and Rui Wang
6. Self-consistent theory to describe charge inversion and like-charge attraction in multivalent electrolytes
 2023 American Physical Society March Meeting, Oral Presentation
Nikhil R. Agrawal and Rui Wang
7. A self-consistent theory for complex electrostatic phenomena at interfaces
 2022 American Chemical Society Colloid and Surface Science Symposium, Oral Presentation
Nikhil R. Agrawal and Rui Wang
8. Beyond mean-field Poisson-Boltzmann: A self-consistent theory for electrical double layers
 Formulation and Material Science Division, Dow Chemicals, Invited Presentation, 2022
Nikhil R. Agrawal

COURSEWORK AND CERTIFICATIONS

- Process Engineering for Silicon wafers (Lam Research)
- Process Integration for Semiconductor manufacturing (Lam Research)
- Electroplating for Semiconductor manufacturing (Lam Research)
- Deposition and Etch for Semiconductor manufacturing (Lam Research)
- Fundamentals of Deep Learning (NVIDIA)
- Data Parallelism: How to Train Deep Learning Models on Multiple GPUs (NVIDIA)
- Model Parallelism: Building and Deploying Large Neural Networks (NVIDIA)
- Bayesian Data Analysis and Machine Learning for Physical Sciences (UC Berkeley)
- Machine Learning, Statistical Models, and Optimization for Physical Sciences (UC Berkeley)

AWARDS AND HONORS

- Langmuir Graduate Student Oral Presentation Awards Finalist, American Chemical Society 2022
- *Institute Silver Medal* from IIT Delhi for highest GPA in the dual degree program in Chemical Engg. 2018
- IIT Delhi *Semester Merit Award* for 9 out of 10 semesters for being among *top 7%* meritorious students across the dual degree program, *consecutively for 8 semesters* 2013-2017
- *Significant Contribution to Research Activities Award* by Chemical Engineering Society, IIT Delhi 2017
- *Kishore Vaigyanik Protsahan Yojana (KVPY) Fellowship* by Dept. of Science and Technology, India 2013
- Was among the *Nation's Top 1%* merit holders in National Standard Examination in Physics (NSEP) conducted by the Indian Association of Physics Teachers (IAPT) 2013

LEADERSHIP ROLES

Secretary, SPIC MACAY, BRCA, IIT Delhi 2015 - 2016

SPIC MACAY: *Society for Promotion of Indian Classical Music And Culture Amongst Youth*

One of the 11 club secretaries of the Board of Recreational and Cultural Activities (BRCA), IIT Delhi.

- Led a team of 13, one each from every hostel in IIT Delhi, to organize marketing, publicity, and hospitality of club events like dance workshops and music concerts
- Successfully secured a sponsorship of Rs. 1 lac from the Gas Authority of India Ltd. (GAIL) to organize *Saaz*, the classical music night of Rendezvous 2015 (IIT Delhi's annual cultural festival), where Grammy Award winner Pt. Vishwa Mohan Bhatt performed.
- Overall coordinator of SPIC MACAY's VIRASAT '16, annual classical music festival of IIT Delhi; headed a team of 15 to organize a 2 week long series of cultural events

Event Coordinator, IIT Delhi Alumni Association April '16

Lead a team of student volunteers for audience management in a seminar on "Ethics and Happiness" by *Nobel Laureate 14th Dalai Lama*

MISCELLANEOUS

Programming Skills

Technical softwares

Material Characterization

Other Interests

Python, C++, PyTorch, GPyTorch, TensorFlow, Scikit Learn, Bash
JMP, MATLAB, Aspen, COMSOL, FLUENT, ParaView, Autodesk Inventor
XSEM, STEM, SIMS, XRR, FTIR, CV/IV (Hg Probe).
Sociology, Reading about emerging technologies and their impact on society